

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

Answer

- A. **6**
- B. **10**
- C. **14**
- D. **24**

Correct Answer: D

Rationale

Choice D is correct. The perimeter, P , of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of **12** centimeters. Substituting **12** for s in the formula for the perimeter of a square yields $P = 4(12)$, or $P = 48$. Therefore, the perimeter of square X is **48** centimeters. It's also given that the perimeter of square Y is **2** times the perimeter of square X. Therefore, the perimeter of square Y is **2(48)**, or **96**, centimeters. Substituting **96** for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by **4** gives $24 = s$. Therefore, the length of one side of square Y is **24** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.